

🌿 SWACHH BHARAT MISSION 2.0

⚡ SMART CITIES MISSION

🇮🇳 DIGITAL INDIA • UPI RAILS

🎯 UN SDGS 11, 12, 13

PROJECT PROPOSAL • SMART CIVIC TECH & CIRCULAR ECONOMY

ECO: AI-Powered Smart Waste Management & Citizen Reward Platform

A full-stack, vision-AI driven civic recycling ecosystem converting domestic unsegregated waste into verifiable digital revenue and real-time municipal intelligence.



DOMAIN THEME	CORE AI MODEL	FINTech INTEGRATION	PRIMARY BENEFICIARY
Waste-to-Wealth / Civic IoT	Gemini 1.5 Flash Vision	Direct UPI Instant Payouts	Households, ULBs & Drivers

📄 Executive Summary

Rapid urban expansion in India produces over 150,000 metric tons of municipal solid waste daily, with less than 25% scientifically segregated at source. **ECO (EcoReward)** solves this crisis by combining **Multimodal Artificial Intelligence (Google Gemini 1.5 Flash Vision)**, **dynamic route-optimized logistics**, and **direct UPI financial micro-incentives**. Citizens scan recyclables using their smartphone camera to verify purity and book doorstep pickups. In return, they receive liquid EcoPoints directly convertible into their bank accounts via UPI (2 pts = ₹1). Meanwhile, urban local bodies (ULBs) and sanitation departments gain live waste heatmaps, illegal dumping resolution workflows, and verifiable ESG carbon accounting.

01 The Problem Statement & National Imperative

72%
UNSEGREGATED AT
SOURCE

₹14,000 Cr
ANNUAL LANDFILL LOSS

32%
COLLECTION FUEL WASTE

4.5M
INFORMAL WASTE PICKERS

Current civic waste handling suffers from a triple failure: (1) **Lack of Source Incentives** (citizens have zero financial motivation to clean and segregate recyclables), (2) **Inefficient Static Fleet Routing** (garbage trucks burn excessive diesel without real-time volume sensing), and (3) **Absence of Traceability** (municipalities cannot identify illegal blackspots or verify recycling quotas).

02 Closed-Loop System Architecture



STEP 01 • CITIZEN

Vision AI Scan

User captures waste image; Gemini 1.5 Flash classifies material (PET, Cardboard, Can) & purity score.

STEP 02 • LOGISTICS

Smart Route Dispatch

Driver receives dynamic sequence sorted by proximity & high-priority bulk weight, reducing fuel by 28%.

STEP 03 • AUDIT

Doorstep Verification

Driver arrives, weighs recyclables, verifies AI estimate, and confirms pickup with digital quality gate.

STEP 04 • REWARD

Instant UPI Cashout

EcoPoints immediately credited to EcoWallet. Citizen redeems direct rupees to Bank VPA via UPI or EcoStore.

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Multi–Stakeholder Ecosystem Architecture

 **Citizen & Household App**

Frontend Web/PWA

- Doorstep Pickup Scheduler:** Multi-material pickup bookings (Plastic, Paper, Metal, Glass, E-Waste).
- Real-time AI Camera:** Purity grading, anti-fraud image classification, and instant point estimation.
- UPI Direct Cashout:** Instant bank payout (2 EcoPoints = ₹1) directly through PhonePe, GPay, Paytm.
- EcoStore & Coupons:** Zero-waste products, grocery vouchers, and food brand coupons.
- Geo-tagged Dump Reporter:** One-tap blackspot camera reporting to the municipal grievance cell.

 **Driver & Logistics Engine**

Field Logistics

- AI Route Sequencing:** Dynamic shortest-path navigation prioritizing bulk recyclables.
- Digital Gate Pass:** QR code check-in at municipal MRF (Material Recovery Facility) yards.
- Quality Audit Screen:** Field weight adjustment tool with anti-tamper confirmation.
- Driver Telematics:** Daily earnings, EV battery health, active shifts, and hazard road alerts.

 **Municipality & ULB Portal**

Civic Governance

- Ward-level Waste Heatmap:** Visual GIS representation of waste generation & segregation compliance.
- Grievance Redressal Cell:** Triage citizen-reported illegal dumping blackspots with before/after photos.
- Landfill Diversion Telemetry:** Verifiable tonnage diverted from landfills into circular economy channels.

 **Central Governance & ESG**

Master Admin

- Automated Carbon Offset Certificate:** Computes CO2 averted and trees saved with verifiable QR links.
- Corporate ESG Compliance:** Real-time sustainability metrics for apartments & corporate offices.
- Fraud Prevention & KYC:** Real-time monitoring of user submissions and driver license verification.

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Computer Vision AI Pipeline & Prompt Architecture

ECO connects directly to **Google Gemini 1.5 Flash Vision API**. Images submitted by citizens are validated through a zero-shot multi-parameter classification prompt:

```
POST /api/advanced/ai/scan-waste | Model: gemini-1.5-flash:generateContent { "wasteType": "Plastic", "materialSubtype": "PET Bottle #1", "confidenceScore": 0.96, "estimatedWeight": 1.5, "qualityScore": 92, "contaminationPct": 5, "remarks": "High-purity PET plastic bottle scan verified with 96% confidence." }
```

05 Technical Specifications & Tech Stack

ARCHITECTURAL LAYER	TECHNOLOGY DEPLOYED	DESIGN RATIONALE
Frontend Architecture	React 18, Vite, Glassmorphism CSS, Lucide Icons	Sub-second load times, mobile-responsive layout for both low-cost smartphones & desktops.
Backend API Core	Node.js, Express.js (REST & Serverless)	Lightweight, asynchronous event-driven I/O handling concurrent pickup requests seamlessly.
Database Engine	MongoDB Atlas (Distributed NoSQL)	Flexible document schema allowing geospatial geoJSON queries for bin locators & heatmaps.
Artificial Intelligence	Google Gemini 1.5 Flash Vision	Multimodal image reasoning identifying 6 major material classes & contamination factors.
Payment Rails	Direct UPI Virtual Payment Address (VPA)	Zero-friction instant micro-credits directly to beneficiaries without intermediary wallet lock-ins.

06 Socio-Economic Impact & National Alignment

 Ecological Footprint Aversion: Recycling 1 ton of segregated plastic conserves 5,774 kWh of electricity, 16.3 barrels of petroleum, and prevents 2.8 tons of atmospheric CO2 emissions.	 Informal Worker Empowerment: Sanitation drivers receive organized schedules, transparent digital wages, anti-hazardous material warnings, and digital gate pass access.
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07 Alignment with Vishwakarma & National Missions

NATIONAL MISSION	TARGET INITIATIVE	ECO PLATFORM CONTRIBUTION
Swachh Bharat 2.0	Garbage-Free Cities (GFC) & Source Segregation	Direct doorstep recycling monetization yielding >85% compliant household segregation.
Digital India	UPI Financial Inclusion for All	Direct micro-transfers of waste scrap value directly into citizen bank accounts.
Mission LiFE	Pro-Planet People (Lifestyle for Environment)	Gamified citizen leaderboards, neighborhood eco-challenges, and carbon offset certificates.

 Conclusion & Future Scalability: ECO provides a turnkey, production-ready blueprint that unites citizens, sanitation workers, and city administration into a self-sustaining circular economy. Future milestones include ultrasonic IoT bin sensor integration and Reverse Vending Machine (RVM) hardware automation.
